

Lecture 21: Vector Fields

Background & Motivation

We now begin the final chapter of the course: vector calculus. Until now, our functions have been scalar-valued— $f(x, y)$ returns a number. A vector field $\mathbf{F}(x, y)$ returns a vector at each point: think wind velocity, magnetic force, or fluid flow. Over the next six lectures, we'll integrate vector fields along curves (line integrals), across surfaces (flux integrals), and connect everything through the great theorems of Green, Stokes, and Gauss. These theorems unify all of calculus into a single principle: the integral of a derivative over a region equals the integral of the function over the boundary.

1. Warm-Up: Gradient Review

Find ∇f for $f(x, y, z) = x^2y + yz^3$.

1. $f_x = 2xy$
2. $f_y = x^2 + z^3$
3. $f_z = 3yz^2$
4. $\nabla f = \langle 2xy, x^2 + z^3, 3yz^2 \rangle$

– The gradient assigns a *vector* to each point – this is an example of a **vector field**.

2. Vector Field Definition

Definition 1: Vector Field

A **vector field** on $\mathbb{R}^2$ is a function $\mathbf{F}(x, y) = P(x, y)\hat{\mathbf{i}} + Q(x, y)\hat{\mathbf{j}}$.

A **vector field** on $\mathbb{R}^3$ is $\mathbf{F}(x, y, z) = P\hat{\mathbf{i}} + Q\hat{\mathbf{j}} + R\hat{\mathbf{k}}$.

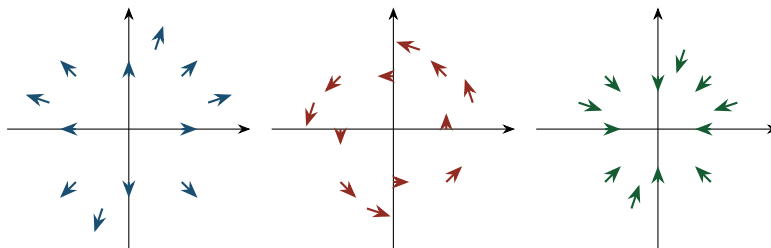
- At each point, $\mathbf{F}$ assigns a vector (direction and magnitude)
- Examples: wind velocity, gravitational force, electric field, fluid flow

3. How to Draw a Vector Field

Procedure for Sketching $\mathbf{F}(x, y)$

1. Pick a grid of sample points (e.g., $(\pm 1, 0)$, $(0, \pm 1)$, $(\pm 1, \pm 1)$, etc.)
 2. At each point (a, b) : evaluate $\mathbf{F}(a, b) = \langle P(a, b), Q(a, b) \rangle$
 3. Draw an arrow **starting at** (a, b) , pointing in the direction $\mathbf{F}(a, b)$
 4. Arrow length $\propto \|\mathbf{F}(a, b)\|$ (longer arrow = stronger field)
- **Key:** the arrow's *tail* is at the sample point, not its midpoint
 - **Normalization:** $\hat{\mathbf{F}} = \mathbf{F}/\|\mathbf{F}\|$ makes all arrows length 1 (shows direction only)

Common Patterns



Source: $\mathbf{F} = \langle x, y \rangle$ **Rotation:** $\mathbf{F} = \langle -y, x \rangle$ **Sink:** $\mathbf{F} = \langle -x, -y \rangle$

- **Source:** vectors point away from origin – fluid is being “created”
- **Sink:** vectors point toward origin – fluid is being “absorbed”
- **Rotation:** vectors swirl around a point – “whirlpool” pattern

4. Examples

Example 1: Evaluating and Sketching a Rotation Field

Let $\mathbf{F}(x, y) = \langle -y, x \rangle$. Evaluate at several points and describe the pattern.

1. **Evaluate at sample points:**

(a, b)	$\mathbf{F}(a, b) = \langle -b, a \rangle$	Direction
$(1, 0)$	$\langle 0, 1 \rangle$	up
$(0, 1)$	$\langle -1, 0 \rangle$	left
$(-1, 0)$	$\langle 0, -1 \rangle$	down
$(0, -1)$	$\langle 1, 0 \rangle$	right
$(1, 1)$	$\langle -1, 1 \rangle$	upper-left
$(2, 0)$	$\langle 0, 2 \rangle$	up (longer)

2. **Sketch:** At each point, draw arrow *starting at that point* in the direction computed.

3. **Pattern:** Vectors circulate **counterclockwise** around the origin.

Magnitude: $\|\mathbf{F}\| = \sqrt{x^2 + y^2} = r$ – arrows get longer farther from the origin.

This is a **rotation field** (also called a **vortex**).

Example 2: Evaluating a Source Field

Let $\mathbf{F}(x, y) = \langle x, y \rangle$. Evaluate at $(1, 0)$, $(0, 2)$, $(-1, 1)$.

1. $\mathbf{F}(1, 0) = \langle 1, 0 \rangle$ – points right from $(1, 0)$

2. $\mathbf{F}(0, 2) = \langle 0, 2 \rangle$ – points up from $(0, 2)$
3. $\mathbf{F}(-1, 1) = \langle -1, 1 \rangle$ – points upper-left from $(-1, 1)$
4. **Pattern:** Every vector points **radially outward** from the origin. This is a **source field**.
Magnitude $= \sqrt{x^2 + y^2} = r$; arrows grow with distance.

5. Conservative Vector Fields

Definition 2: Conservative Field

$\mathbf{F}$ is **conservative** if there exists a scalar function f (the **potential function**) such that:

$$\mathbf{F} = \nabla f$$

- Work done by a conservative field is **path-independent**
- In physics: conservative $\iff$ energy is conserved (e.g., gravity, electrostatics)

Theorem 1: Test for Conservative Fields (2D)

$\mathbf{F} = \langle P, Q \rangle$ is conservative on a simply connected domain if and only if:

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$$

3D version: $\mathbf{F} = \langle P, Q, R \rangle$ is conservative iff $\nabla \times \mathbf{F} = \mathbf{0}$:

$$P_y = Q_x, \quad P_z = R_x, \quad Q_z = R_y$$

Caution: The domain must be **simply connected** (no holes) for the test to be sufficient.

Simply Connected Domains

- **Simply connected** – every closed curve can be shrunk to a point without leaving the domain
- $\mathbb{R}^2$ and $\mathbb{R}^3$: simply connected ✓
- $\mathbb{R}^2 \setminus \{(0, 0)\}$: **not** simply connected (has a “hole”)

6. Finding Potential Functions

Algorithm for Finding f (2D)

Given $\mathbf{F} = \langle P, Q \rangle$ that passes the conservative test:

1. **Integrate:** $f = \int P \, dx$ (treat y as constant) $\implies f = (\dots) + g(y)$
2. **Differentiate:** $f_y = (\dots) + g'(y)$. Set equal to Q and solve for $g'(y)$.
3. **Integrate** $g'(y)$ to find $g(y)$. Combine.

For 3D: same idea but with $g(y, z)$, then $h(z)$.

Example 3: Testing and Finding a Potential Function

Is $\mathbf{F} = \langle 2xy + y^2 \cos x, x^2 + 2y \sin x \rangle$ conservative? If so, find f .

1. **Test:** $P = 2xy + y^2 \cos x$, $Q = x^2 + 2y \sin x$.

$$P_y = 2x + 2y \cos x$$

$$Q_x = 2x + 2y \cos x$$

$P_y = Q_x$ ✓ – Conservative!

2. **Integrate P with respect to x :**

$$f = \int (2xy + y^2 \cos x) dx = x^2 y + y^2 \sin x + g(y)$$

3. **Differentiate with respect to y and match to Q :**

$$f_y = x^2 + 2y \sin x + g'(y) = Q = x^2 + 2y \sin x$$

$$\implies g'(y) = 0 \implies g(y) = C$$

4. **Result:** $f(x, y) = x^2 y + y^2 \sin x + C$

Example 4: Non-Conservative Field: The Hole Counterexample

Show that $\mathbf{F} = \left\langle \frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right\rangle$ passes $P_y = Q_x$ but is NOT conservative.

1. **Check partials:**

$$P_y = \frac{y^2 - x^2}{(x^2 + y^2)^2}, \quad Q_x = \frac{y^2 - x^2}{(x^2 + y^2)^2}$$

$P_y = Q_x$ ✓ – test satisfied!

2. **But the domain is $\mathbb{R}^2 \setminus \{(0, 0)\}$** – not simply connected (hole at origin).

3. **Proof:** Integrate around the unit circle C : $x = \cos t$, $y = \sin t$, $0 \leq t \leq 2\pi$:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} (\sin^2 t + \cos^2 t) dt = 2\pi \neq 0$$

If $\mathbf{F}$ were conservative, every closed-loop integral would be 0. Contradiction!

4. **Lesson:** Simply connected domain is **essential** for the $P_y = Q_x$ test.

Practice Problems

Problem 1. Evaluate $\mathbf{F}(x, y) = \langle x^2, xy \rangle$ at $(1, 2)$, $(2, 1)$, $(-1, 3)$. For each, state where the arrow starts and what direction it points.

$\mathbf{F}(1, 2) = \langle 1, 2 \rangle$: arrow starts at $(1, 2)$, points in direction $\langle 1, 2 \rangle$ (up-right, steep).

$\mathbf{F}(2, 1) = \langle 4, 2 \rangle$: arrow starts at $(2, 1)$, points in direction $\langle 4, 2 \rangle$ (right-ish, shallow).

$\mathbf{F}(-1, 3) = \langle 1, -3 \rangle$: arrow starts at $(-1, 3)$, points in direction $\langle 1, -3 \rangle$ (down-right).

Problem 2. Determine if $\mathbf{F} = \langle ye^{xy}, xe^{xy} \rangle$ is conservative. If so, find f .

1. **Test:** $P_y = e^{xy} + xye^{xy}$, $Q_x = e^{xy} + xye^{xy}$. Equal ✓

2. **Integrate:** $f = \int ye^{xy} dx = e^{xy} + g(y)$

3. **Match:** $f_y = xe^{xy} + g'(y) = xe^{xy} \implies g'(y) = 0$

4. **Result:** $f = e^{xy} + C$

Problem 3. Show that $\mathbf{F} = \langle x^2 - y^2, 2xy \rangle$ is NOT conservative.

$$P_y = -2y, \quad Q_x = 2y.$$

Since $-2y \neq 2y$ (for $y \neq 0$), $P_y \neq Q_x$. NOT conservative.

Problem 4. Find the potential function for $\mathbf{F} = \langle 2xz + y, x + z^2, x^2 + 2yz \rangle$.

1. **Integrate P w.r.t. x :** $f = \int (2xz + y) dx = x^2z + xy + g(y, z)$

2. **Match $f_y = Q$:** $x + g_y = x + z^2 \implies g_y = z^2 \implies g = yz^2 + h(z)$

3. **Match $f_z = R$:** $x^2 + 2yz + h'(z) = x^2 + 2yz \implies h'(z) = 0$

4. **Result:** $f = x^2z + xy + yz^2 + C$

Problem 5. Let $\mathbf{F} = \langle y \cos(xy) + 1, x \cos(xy) + 2z, 2y + 3z^2 \rangle$.

(a) Verify $\mathbf{F}$ is conservative (check all three conditions).

(b) Find the potential function.

(a) Test:

$$P_y = \cos(xy) - xy \sin(xy), \quad Q_x = \cos(xy) - xy \sin(xy) \quad \checkmark$$

$$Q_z = 2, \quad R_y = 2 \quad \checkmark$$

$$P_z = 0, \quad R_x = 0 \quad \checkmark$$

All three match – conservative.

(b) Find f :

1. $f = \int (y \cos(xy) + 1) dx = \sin(xy) + x + g(y, z)$

2. $f_y = x \cos(xy) + g_y = x \cos(xy) + 2z \implies g_y = 2z \implies g = 2yz + h(z)$

3. $f_z = 2y + h'(z) = 2y + 3z^2 \implies h'(z) = 3z^2 \implies h = z^3$

4. **Result:** $f = \sin(xy) + x + 2yz + z^3 + C$